Alan Tseng

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SUMMARY

Embedded platform engineer who brings up silicon at the register level – device tree, I2C and PMBus drivers, bit-field decoding against vendor datasheets, cross-debugging on the target – and builds AI agents to automate that work. The same workflow carries from server management controllers (OpenBMC device enablement and firmware debugging on ASPEED BMCs) through to Android SoC/AP platform integration. Triage firmware crashes with gdbserver and core dump analysis, profile with perf and uftrace, and deliver against datacenter customer specifications. M.Sc. from the Graduate Institute of Networking and Multimedia, National Taiwan University. COSCUP speaker on OpenBMC debugging (2025) and agentic AI for firmware development (2026).

SKILLS

Languages	C, C++, Python, Shell, JavaScript / TypeScript
Systems & Kernel	firmware / core dump debugging, Android AOSP, device tree, drivers (I2C, GPIO, ADC), systemd, D-Bus, performance analysis (perf, ftrace, uftrace), GDB / gdbserver, core dump analysis
Firmware & Embedded	OpenBMC, Yocto, Buildroot, IPMI, Redfish, BMC firmware development, sensor enablement
AI & Automation	LLM agent frameworks, RAG (embedding + vector store retrieval), tool-use / function calling, agent skill & tool-use workflow design
Infrastructure & Cloud	Kubernetes (kubeadm, Helm), Docker, PXE / DHCP / TFTP provisioning, CI/CD pipelines, image / provisioning footprint optimization
Web & Tooling	WebGPU, WebAssembly (WASM), Web APIs, browser-based benchmarking, REST API design, pytest

EXPERIENCE

Project System Integration Engineer – Android Platform Sep. 2025 – Present
MediaTek, Hsinchu, Taiwan

- As owner of a driver module on a commercial tablet program, systematized the new-chip porting workflow – config, device tree nodes, register base updates, build wiring – into an internal AI agent skill, making repeat ports reproducible.
- Wrote a set of internal AI agent skills that automate repetitive Android platform work: a system-upgrade skill that pulls target changes from Gerrit and runs local verification, and a build-and-boot skill that fetches the code, builds the image, flashes the device, and confirms it boots to the home screen.
- Won an internal engineering award for an agent toolset that lets teams adopt AI agents without exposing internal source or credentials; designed its defense-in-depth architecture – reasoning and action separated under least-privilege and complete-mediation – and built the namespace isolation between the two layers as a proof-of-concept. Published the architecture as a public writeup.
- Prototyped a Tauri-based desktop tool (Python backend driving Claude Code over a pty) that integrates internal Teams, Outlook, and Jira with an in-house AI inference backend to automate code-owner notification, removing a manual coordination step.
- As task leader, benchmarked a competitor commercial tablet against internal silicon with no developer access to it; wrote custom WebGPU / WebAssembly / Web API programs to measure CPU and GPU performance on locked production hardware.
- As task leader and software coordinator, designed the performance and low-power evaluation methodology and coordinated QA and SA execution, including customer-board rework; consolidated results into reports that fed product and platform decisions.
- Maintain branch, repo, CI, and image pipelines for the program; root-cause build failures, runtime logs, and benchmark regressions across the vendor and AOSP system layers, and drive fixes across software and QA teams.

Embedded Systems Engineer – BMC Firmware Dec. 2024 – Sep. 2025
Wiwynn, New Taipei City, Taiwan

- Developed and debugged OpenBMC firmware on Yocto and Buildroot hyperscale server platforms, working on sensors, services, and management interfaces.
- Developed device-specific handling in the PMBus driver path – I2C/SMBus register access and bit-field decoding per vendor datasheet – to expose correct hwmon/sysfs sensor values for datacenter customer platform requirements.
- Brought up and debugged device tree, I2C, GPIO, ADC, systemd, and D-Bus interfaces; resolved issues on IPMI and Redfish management endpoints against datacenter customer platform requirements.
- Built a cross-debugging workflow with gdbserver, core dump analysis, and live-attach to triage BMC firmware crashes and service failures on production servers. Presented it as a public talk at COSCUP 2025.
- Wrote Python and pytest validation tools that automated manual sanity checks before customer hand-off.

Server Software Engineer – Provisioning Automation Sep. 2024 – Dec. 2024
Pegatron, Taipei City, Taiwan

- Designed and built Kubernetes-based PXE provisioning for rack-scale ARM and x86 server validation, automating Windows and Linux OS deployment with kubeadm, Helm, Docker, DHCP, and TFTP.
- Wired low-level networking (ARP, DHCP, TFTP) into the pipeline and added stage-by-stage logging so boot failures could be diagnosed without a physical console.
- Optimized the provisioning architecture to cut image and repository footprint from 663 MB to 2.4 MB (about 99%) by moving the bundled `.deb` packages and program artifacts out of the image and serving them from a file server on the Kubernetes node instead of baking them in.

Software Engineer Intern Jul. 2022 – Nov. 2022
LINE Corporation, Taipei City, Taiwan

- Built and deployed a PoC that auto-generates a mock server from the team's OpenAPI spec, letting the frontend team start parallel development without hand-writing mocks; standardized API specs across the B2B platform team.

EDUCATION

M.Sc., Graduate Institute of Networking and Multimedia, College of Electrical Engineering & Computer Science, National Taiwan University Aug. 2021 – Feb. 2024 | GPA: 4.12/4.3
B.S., Computer and Communication Engineering, Ming Chuan University Sep. 2017 – Jun. 2021 | GPA: 3.99/4.0

SELECTED PROJECTS

- **CPU-Side Low-Bit LLM Inference Analysis (T-MAC / BitNet)** – Profiled inference performance using `perf` and `uftrace`; analyzed IPC, cache and TLB misses, page faults, and function-level execution behavior to identify optimization targets.
- **Android Common Kernel Contribution** – Patch merged into `kernel/common` on `android-mainline`, restoring the `struct dma_heap` kernel-doc lost in a merge. gerrit
- **Open-Source Translation & Course Materials** – Translated *Operating System in 1000 Lines* into Traditional Chinese – all 21 chapters, now the project's zh-TW locale. Wrote Linux kernel course materials on eBPF observability, perf and flamegraph profiling, and memory management. translation | wiki
- **Adaptive Driving Beam System** – Designed a nighttime vehicle-detection and adaptive-beam system using CycleGAN, YOLO, statistical analysis, and SVM; published in CVGIP 2020.
- **Technical Writing** – Maintain a TIL and blog covering embedded systems, Linux debugging, AI-assisted development, and platform engineering workflows. TIL | Blog

TALKS, HONORS AND LEADERSHIP

- **Talk** – *Agentic AI for Firmware Development*, COSCUP 2026. How LLM agents fit into firmware and embedded work: understanding code, assisting debugging, and using tools safely inside engineering pipelines.
- **Talk** – *Debugging OpenBMC Firmware*, COSCUP 2025. Cross-debugging on production BMC platforms with gdbserver, core dump analysis, and live-attach.
- **Honors** – Research Excellence Award (NTU); Intel Andy Grove Scholarship; College Student Research Scholarship from Taiwan's Ministry of Science and Technology.
- **Leadership** – Lead, Google Developer Student Club; volunteer staff at SITCON and COSCUP community events.